

Hybrid Approach for Machine Anomaly Detection Across Diverse Operating Conditions: Combining Self-Supervised Learning With Vibration Statistics

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Abstract—Failure detection in rotating machines is critical for reducing maintenance costs, economic losses, and safety-related accidents. Deep learning has become a robust and powerful solution for machine fault diagnosis, but it faces two major challenges: scarcity of labeled faulty data and variability of machine operating conditions. To address these challenges, we propose a novel hybrid machine anomaly detection methodology that integrates an innovative pretext task-based self-supervised learning framework with vibration statistics. First, we designed a pretext task to classify the rotating speeds and trained an auxiliary classifier on it. This auxiliary classifier effectively extracts fault-discriminative features related to rotating speed, an operating condition closely associated with various machine faults. Moreover, the proposed pretext task applies to the fault diagnosis of general rotating machines because the rotating speed can be easily obtained using a Hall sensor. Second, a hybrid feature was generated by combining the auxiliary classifier outputs with six statistical features derived from raw vibration signals. These statistical features augment the fault-discriminative ability of the auxiliary classifier outputs by incorporating the magnitude and asymmetry of the signal, which the auxiliary classifier alone cannot capture. The evaluation results demonstrated that our approach significantly outperformed 11 state-of-the-art anomaly detection methods on three different vibration datasets encompassing multiple operating conditions and a real-world run-to-failure scenario. In addition, our method is capable of performing real-time inference on an edge device platform, demonstrating its applicability to practical industrial setups. Furthermore, feature visualization analysis highlighted how the proposed hybrid feature enables a more distinct separation between faulty and healthy data samples than that achieved with statistical features or auxiliary classifier outputs alone.

Index Terms—Anomaly detection, diverse operating conditions, feature integration, machine fault diagnosis, self-supervised learning, vibration statistics.

I. INTRODUCTION

MECHANICAL system failures can result in increased maintenance costs, diminished operational efficiency, and safety-related accidents. The prompt and accurate diagnosis of machine faults is crucial for minimizing both economic

and human losses in industrial environments. Rotating machinery, a vital component of modern industry, is extensively employed in various systems such as railways, aircraft, and turbines. Therefore, faults in rotating machinery are often a primary cause of overall mechanical system failures.

Vibration is one of the most widely utilized physical parameters for assessing the condition of rotating machinery [1]. Traditional vibration-based machine fault diagnosis methods usually rely on frequency analysis techniques, which analyze the fault-related characteristics in the frequency domain of the signal to examine the machine condition [2], [3]. However, these approaches typically require expert engineers with extensive domain knowledge to make diagnostic decisions, which limits their scalability. Recently, data-driven approaches, such as machine learning and deep learning, have been actively investigated, which automatically learn fault-related features from the data to diagnose machine faults. In particular, recent studies have focused on developing deep learning model architectures, such as convolutional neural networks (CNNs) [4], [5], recurrent neural networks (RNNs) [6], and Transformers [7], [8], to automatically extract high-level features from raw vibration signals.

Despite their potential, the application of data-driven fault diagnosis methods in real-world industrial environments faces two major challenges. First, a large volume of labeled faulty data is often unavailable [9], [10]. Although a substantial amount of labeled faulty data is essential for improving the diagnostic performance of models, faults in industrial machinery are rare, as the machines typically operate under healthy conditions. In addition, labeling the presence and type of faults requires skillful engineers' annotation, which incurs significant costs. Consequently, developing diagnostic methods that rely only on healthy data is crucial. Second, industrial machinery operates under various conditions, such as varying rotating speeds, motor loads, and mechanical components [11]. This variability causes shifts in the data distribution and increases the data complexity, which degrades the diagnostic performance of the model. As a result, it becomes challenging to generalize the diagnostic capability of the model across different operating conditions.

To address the first challenge, anomaly detection has emerged as a viable solution. This method involves training models on normal data instances to detect anomalies, also

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referred to as outliers, which deviate from the patterns of normal instances [12], [13]. In machine fault diagnosis, anomaly detection has been frequently applied by treating healthy samples as normal and faulty samples as anomalies.

Research on machine anomaly detection can be classified into three categories: 1) machine learning-based methods; 2) reconstruction methods; and 3) self-supervised learning-based methods. Machine learning-based methods detect anomalies using statistical models [14], [15], [16], which rely on manually engineered features. This approach requires a high level of domain expertise to effectively extract the fault-discriminative features. Reconstruction methods identify anomalies based on the reconstruction errors of the autoencoder (AE) architectures, which compress and reconstruct the data [17]. Recently, reconstruction methods have been extended to include adversarial training strategies [18], [19] or combined with machine learning-based techniques [20]. However, they often experience significant performance degradation when dealing with complex distributions of healthy data [12]. Consequently, applying these methods to modern industrial environments with diverse operating conditions remains challenging.

Self-supervised learning-based methods detect anomalies by leveraging the information intrinsic to the data, eliminating the need for a labeling process [21]. This approach has attracted considerable attention because it allows supervised learning techniques to be applied to unlabeled data. In self-supervised learning, a learning task used for model training determines the features that the model extracts. Therefore, the fault-discriminative ability of the model is highly dependent on the design of the learning task. Although several studies have applied self-supervised learning to machine anomaly detection, research on designing the learning task specifically tailored to rotating machinery fault diagnosis under diverse operating conditions remains scarce.

To overcome the limitations of previous machine anomaly detection methods, this paper proposes a novel hybrid approach for machine anomaly detection across various operating conditions. The proposed approach consists of pretext task-based self-supervised learning to train an auxiliary classifier and integration of the features extracted from the auxiliary classifier and statistical parameters. The main contributions of this study are as follows.

- 1) This study addresses fault diagnosis in rotating machines under diverse operating conditions without requiring labeled faulty data samples. Unlike previous studies that primarily assumed fixed operating conditions, our approach effectively handles variations in multiple operational states, including those related to rotating speed, motor load, and bearing type.
- 2) We introduce a novel self-supervised learning framework, which is broadly applicable to the fault diagnosis of rotating machinery. In this framework, we propose a pretext task to classify the readily available rotating speed categories. Rotating speed is a key factor in capturing the fault characteristics. Moreover, it can be easily obtained for all types of rotating machines by using Hall sensors. Therefore, the proposed learning

task ensures both general applicability and strong fault-discriminative capability.

- 3) To enhance the feature extraction capability of the auxiliary classifier, we adopt a hybrid approach that combines features extracted from the auxiliary classifier with vibration statistics. This integration alleviates the limitations of the statistical features and auxiliary classifier outputs, enabling our method to effectively distinguish between complex fault-related patterns across diverse operating conditions.
- 4) This study demonstrates the practical applicability of the proposed method to real-world industrial settings from two perspectives: anomaly detection under run-to-failure scenarios and real-time inference on an edge device.

The remainder of this paper is organized as follows. Section II reviews previous research on machine anomaly detection and discusses their limitations. Section III describes the proposed anomaly detection methodology. Section IV describes the dataset, experimental setup, and performance evaluation of the proposed method against existing approaches. Section V presents the results of ablation studies and feature visualizations, highlighting the effectiveness of the proposed approach. Finally, Section VI summarizes the key findings and concludes this study.

II. RELATED WORK

A. Machine Learning-Based Methods

Machine learning-based methods have been widely used for anomaly detection even before the advent of deep learning. Recent studies have focused on improving the feature extraction process. de Novaes Pires Leite et al. [22] proposed a method for diagnosing wind turbine faults by extracting features from the time, frequency, and mechanical domains of vibration signals. Chen et al. [23] investigated a Transformer fault diagnosis method based on impulse frequency response analysis of current signals using a pulse generator. Hu et al. [24] conducted order analysis on the frequency domain of vibration signals to extract vibration vectors for diagnosing faults in rotating machinery.

Machine learning algorithms used in [22], [23], and [24] achieved strong performance in handling manually extracted tabular features. However, they relied on domain-specific analysis for feature extraction. To overcome this limitation, this study employed a machine learning model for anomaly detection but used automatically extracted features without depending on domain-specific decision-making or analysis.

B. Reconstruction Methods

The reconstruction method identifies machine faults by using the reconstruction error of encoder-decoder architectures, such as AEs, as the anomaly score. An AE model consists of an encoder that compresses the input data and a decoder that reconstructs the compressed data to its original form. The model is trained to minimize reconstruction errors, such as the mean squared error, solely with healthy data. Hence, the model accurately reconstructs healthy data but struggles to reconstruct faulty data that are not included in the

training data. Consequently, the reconstruction errors of the faulty data usually exceed those of the healthy data, resulting in data with reconstruction errors exceeding a predefined threshold being classified as faulty.

Many studies have utilized the reconstruction method for machine anomaly detection. Ahmed et al. [25] proposed a gearbox fault diagnosis method using a denoising AE (DAE), and Choi et al. [26] developed a lightweight bearing fault diagnosis model by integrating principal component analysis with AE to reduce the computational demands. Research using AE architectures incorporating CNN or RNN layers has also been conducted. Kim and Ro [27] diagnosed dry vacuum pump faults using features extracted by discrete wavelet transform as the inputs for long short-term memory-based AE. Zhao et al. [28] proposed CAE-CWT, a model that detects bearing faults using a 2-D convolutional AE (CAE) with 2-D spectrograms generated using continuous wavelet transform (CWT). Some studies have employed the emerging Transformer architecture as the encoder-decoder model. For example, Xu et al. [29] proposed an anomaly Transformer (ATF) architecture, which is trained to minimize both the reconstruction error and association discrepancy.

Recently, several studies have incorporated reconstruction methods into adversarial training strategies or combined them with machine learning techniques. In [30], [31], and [32], researchers used generative adversarial networks (GANs) to diagnose faults in bearings, structures, and industrial products, respectively. Hojjati and Armanfard [33] introduced deep autoencoding support vector data descriptor (DASVDD), an anomaly detection method that combines the reconstruction error of AE with the distance metric of SVDD. Yoon and Yu [34] and Du et al. [35] developed methods for machine anomaly detection by combining the AE model with SVDD and isolation forest (iForest), respectively.

Unlike machine learning-based methods, reconstruction methods are well-suited for raw input data, such as vibration and sound signals from industrial environments. However, in reconstruction methods, the model is trained to reconstruct the input data accurately after compression, rather than discriminate faulty data from healthy data. Consequently, its performance can degrade when the distribution of the healthy data used for training is complex [12]. Since this study aimed to address the challenges in fault diagnosis under various operating conditions, reconstruction methods were not utilized. Instead, a self-supervised learning approach was adopted to train a deep learning model as a feature extractor. During the performance evaluation, the proposed approach was compared with representative reconstruction methods across three datasets representing diverse operating conditions.

C. Self-Supervised Learning-Based Methods

Self-supervised learning can be categorized into contrastive learning and auxiliary pretext task learning. Contrastive learning trains a model by using the similarity between data pairs. Specifically, the model is trained to embed similar data pairs, referred to as positive pairs, closer together and dissimilar data pairs, referred to as negative pairs, farther apart in the latent embedding space. Several studies have applied contrastive

learning to machine anomaly detection. Yang et al. [36], Cui et al. [37], and Wang et al. [38] proposed machine anomaly detection methods based on contrastive learning. Yang et al. [36] trained a model using negative sample pairs generated by masking specific regions in the frequency domain of vibration signals. Cui et al. [37] augmented vibration signal data to generate positive samples and trained the model to minimize their distance from the original samples. Wang et al. [38] proposed a self-supervised diagnosis method for faults in industrial processes by utilizing the internal contrastive learning (ICL) strategy [39]. This approach generates positive and negative pairs using only a subset of the training samples, eliminating the need for external data augmentation.

Although contrastive learning has shown effectiveness in machine anomaly detection, it does not fully capture the fault-discriminative characteristics associated with machine operating conditions. Consequently, its performance may degrade under diverse operating conditions. For example, the method in [37] can handle datasets with various motor loads and rotating speeds, but it requires additional labeled faulty data to improve the model performance.

In contrast, auxiliary pretext task learning trains a model through a pretext task based on pseudo-labels generated from metadata. For example, in [40] and [41], models were trained for anomaly detection through a pretext task to classify the machine ID, which is provided as metadata in the DCASE 2020 anomalous sound detection dataset. Since machine ID reflects the operating conditions of machines in the DCASE 2020 dataset, this approach allows the model to effectively handle diverse operating conditions. However, this metadata is specific to the dataset and is not universally applicable to rotating machinery fault diagnosis. To address these limitations in existing pretext task learning-based approaches, this paper presents a new pretext task that can be widely applied to vibration data from rotating machinery operating under various conditions. In summary, the advantage of our approach lies in overcoming the limitations of prior anomaly detection methods based on machine learning and auxiliary pretext task learning.

III. METHODOLOGY

This section describes the proposed method. As illustrated in Fig. 1, the proposed method consists of three components: an auxiliary classifier, a statistical feature extractor, and an anomaly detector. Initially, the auxiliary classifier is trained through a pretext task designed to classify rotating speeds based on vibration signals measured from a healthy machine. During the testing phase, the outputs of the classifier serve as feature vectors for anomaly detection. Next, the statistical feature extractor calculates the statistical features, including the mean (μ), peak (pk), rms, crest factor (CF), skewness (skew), and kurtosis (kurt), from the vibration signals. These statistical features are then combined with the auxiliary classifier outputs to form a hybrid feature. Finally, the anomaly detector identifies machine faults based on the hybrid feature using the local outlier factor (LOF) algorithm.

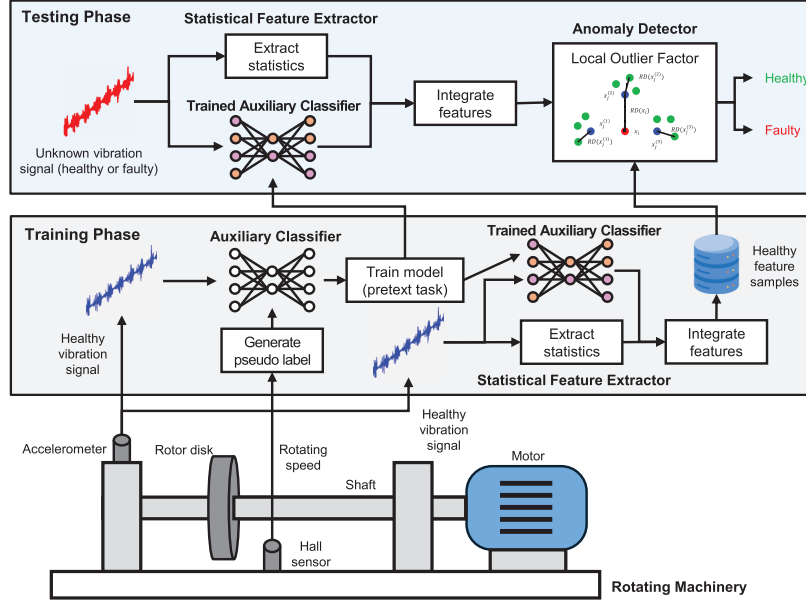


Fig. 1. Schematic architecture of the proposed anomaly detection method.

A. Auxiliary Classifier and Self-Supervised Learning

To distinguish faulty data from healthy data, a pretext task for machine anomaly detection should enable the model to learn fault-related representations of the data from healthy machines under diverse operating conditions. To achieve this goal, pseudo-labels were generated from one-hot encoded rotating speeds, and an auxiliary classifier was trained to classify these labels.

Rotating speed is a physical parameter that can be easily collected from Hall sensors without additional analysis. More importantly, it is a fundamental factor influencing the fault-related behavior of rotating machinery. For example, rotor faults are closely associated with the $\times 1$ and $\times 2$ harmonics of rotating speed. Additionally, bearing fault frequencies, such as ball pass frequency inner (BPFI), ball pass frequency outer (BPFO), and ball spin frequency (BSF), are also directly dependent on the rotating speed. These are defined as

$$\text{BPFI} = r \times (N_B/2) \times (1 + (B_D/P_D) \cos \beta) \quad (1)$$

$$\text{BPFO} = r \times (N_B/2) \times (1 - (B_D/P_D) \cos \beta) \quad (2)$$

$$\text{BSF} = r \times (P_D/B_D) \times (1 - 2 (B_D/P_D) \cos^2 \beta) \quad (3)$$

where r is the rotating speed and N_B , B_D , P_D , and β denote the number of balls, ball diameter, pitch diameter, and contact angle of the bearing, respectively. Therefore, a model trained on a pretext task using the rotating speed information can extract the fault-related features for anomaly detection.

For each data instance x_i , a pseudo-label y_i was assigned in the form of a one-hot vector corresponding to its revolutions per minute (RPM) value. The datasets used in this study, described in Section IV-A, comprise two or more rotating speed conditions. For example, the University of Seoul (UOS) dataset [11] includes data collected at six rotating speeds: 600, 800, 1000, 1200, 1400, and 1600 RPMs. In this case,

Algorithm 1 Training Procedure of the Auxiliary classifier

Require: Auxiliary classifier $f(\cdot) : \mathbb{R}^L \rightarrow \mathbb{R}^C$, model parameter θ .

Ensure: $B \leftarrow$ batch size, $L \leftarrow$ data length, $C \leftarrow$ number of classes.

function UPDATE(f, θ)

Initialize loss $\mathcal{L} = 0$

for $i = 0, \dots, B - 1$ **do**

Get vibration $x \in \mathbb{R}^L$ from accelerometer.

Get rotating speed $r \in \mathbb{R}$ from Hall sensor.

Encode $r \in \mathbb{R}$ to $y \in \{0, 1\}^C$

$z = f(x; \theta) \in \mathbb{R}^C$

$\hat{y} = \text{softmax}(z)$

$\mathcal{L} = \mathcal{L} - \sum_{i=0}^{C-1} y_i \log(\hat{y}_i)$

end for

$\mathcal{L} = \mathcal{L}/B$

Update model parameter θ to $\hat{\theta}$ by backpropagation.

return $\hat{\theta}$

end function

the pseudo-labels for each rotating speed are represented as $y_i \in \{(1, 0, 0, 0, 0, 0), (0, 1, 0, 0, 0, 0), \dots, (0, 0, 0, 0, 0, 1)\}$. Algorithm 1 outlines the training process of the auxiliary classifier. As shown in Table I, the auxiliary classifier $f(\cdot)$ employs a 1-D CNN architecture, which consists of five 1-D convolutional layers, one global average pooling (GAP) layer, and one fully connected (FC) layer.

The i th element z_i in the output $z \in \mathbb{R}^C$ of the trained classifier represents a logit, indicating the likelihood of the input data belonging to the i th rotating speed class. In this study, the auxiliary classifier is trained on healthy data. When a healthy data sample belonging to class i is provided, the classifier typically produces the highest output value, z_i , corresponding to the class i , whereas the outputs for the other classes, $\{z_j | j \neq i\}$, are significantly lower than z_i . By contrast, when a faulty data sample is given, the classifier may produce

TABLE I
ARCHITECTURE OF THE AUXILIARY CLASSIFIER

Layer No.	Type	K	N_c	S	P	Output Size
1	1DConv+BN+ReLU	64	2	2	31	$2 \times L/2$
2	1DConv+BN+ReLU	32	2	2	15	$2 \times L/4$
3	1DConv+BN+ReLU	16	4	2	7	$4 \times L/8$
4	1DConv+BN+ReLU	8	8	2	3	$8 \times L/16$
5	1DConv+BN+ReLU	4	8	2	1	$8 \times L/32$
6	GAP	-	-	-	-	8
7	FC	-	C	-	-	C

1DConv: 1-D convolutional layer, BN: Batch normalization, ReLU: Rectified linear unit, K : Kernel size, N_c : No. channels, C : No. classes, S : Stride, P : Padding, L : Input data length.

TABLE II
SUMMARY OF STATISTICAL FEATURES

Statistics	Notation
Mean (μ)	$\mu(x) = \frac{1}{L} \sum_{i=0}^{L-1} x_i$
Peak (pk)	$pk(x) = \max x_i $
rms	$rms(x) = \sqrt{\sum_{i=0}^{L-1} x_i^2 / L}$
Crest factor (CF)	$CF(x) = pk(x) / rms(x)$
Skewness ($skew$)	$skew(x) = \frac{1}{L} \sum_{i=0}^{L-1} \left(\frac{x_i - \mu(x)}{\sigma(x)} \right)^3$
Kurtosis ($kurt$)	$kurt(x) = \frac{1}{L} \sum_{i=0}^{L-1} \left(\frac{x_i - \mu(x)}{\sigma(x)} \right)^4$

incorrect class predictions, representing patterns that deviate from those observed from healthy data, because faulty data are not included during training. This discrepancy enables the classifier to produce features that effectively distinguish faulty data from healthy data. Consequently, the outputs of the auxiliary classifier are used as a feature vector in our machine anomaly detection method. Further details on how the model outputs distinguish between healthy and faulty data are discussed in Section V-B.

B. Statistical Feature Extractor

The statistical features represent key characteristics of the signal shape, such as the central axis, magnitude, and asymmetry, which are closely associated with fault-related vibration patterns. In this study, six frequently used metrics in vibration analysis [42], listed in Table II, were selected as the statistical features. These metrics are automatically calculated using predefined formulas, eliminating the need for further manual analysis.

The value of μ represents the central axis of the vibration, whereas pk and rms values represent its overall magnitude. The value of CF , calculated as the ratio of pk to rms , indicates the intensity of the impulses within the signal. The value of skew reflects the asymmetry of the data distribution, with large positive and large negative values indicating right-skewed and left-skewed distributions, respectively. The value of $kurt$ quantifies the degree of extreme values in the data. In summary, signals with larger amplitudes exhibit higher values of pk and rms , signals with greater asymmetry exhibit

larger absolute values of μ and skew, and signals with more extreme fluctuations result in higher CF and $kurt$. Faulty machines typically produce large and irregular vibrations, leading to significantly higher absolute values of these metrics. Therefore, these metrics can effectively distinguish between healthy and faulty data.

C. Anomaly Detector

To improve the fault-discriminative capability of the features, a hybrid feature vector was constructed by integrating the outputs from the auxiliary classifier with statistical parameters. Both components independently contribute to fault detection as well as complement each other. The auxiliary classifier effectively extracts the fault-related features associated with rotating speed. However, since its outputs represent the class probabilities, the overall shape of the raw signals may be partially lost during computation. In contrast, statistical features capture the magnitude and shape of the vibration signals, which are closely related to the fault patterns. However, these features are highly sensitive to variations in rotating speed. Consequently, by combining these two feature types, the proposed hybrid feature approach enhances the robustness of anomaly detection across diverse operating conditions.

Formally, for input data $x \in \mathbb{R}^L$ of length L , let the auxiliary classifier output be $z = f(x) \in \mathbb{R}^C$ and the statistical features be $s = [\mu(x), pk(x), rms(x), CF(x), skew(x), kurt(x)] \in \mathbb{R}^6$. The hybrid feature vector $h \in \mathbb{R}^{C+6}$ is defined as

$$h = [z, s]. \quad (4)$$

To prepare the feature for anomaly detection, a hybrid feature set $H = [h_0, h_1, \dots, h_{N-1}]^T \in \mathbb{R}^{N \times (C+6)}$ was generated for N data samples. However, since the features from the auxiliary classifier and statistical parameters have different scales, some features may dominate the anomaly detection process. For example, the auxiliary classifier output z may have varying scales depending on the training results, whereas the statistical feature s is typically measured on an acceleration scale, such as gravitational acceleration. To solve this problem, z score normalization was applied to standardize all features in H with a mean of 0 and standard deviation of 1. The normalized hybrid feature set $\bar{H} \in \mathbb{R}^{N \times (C+6)}$ is calculated as

$$\bar{H}_{i,j} = (H_{i,j} - \mu_H(j)) / \sigma_H(j) \quad (5)$$

$$\mu_H(j) = \frac{1}{N} \sum_{i=0}^{N-1} H_{i,j} \quad (6)$$

$$\sigma_H(j) = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} (H_{i,j} - \mu_H(j))^2} \quad (7)$$

where i and j represent the index of a data sample and a feature, respectively, and $\mu_H(j)$ and $\sigma_H(j)$ are the mean and standard deviation of the j th feature, respectively.

This study aims to address the challenge presented by the diverse operating conditions of rotating machines by using the hybrid feature, which is an automatically extracted high-level feature. As discussed in Section II-B, reconstruction

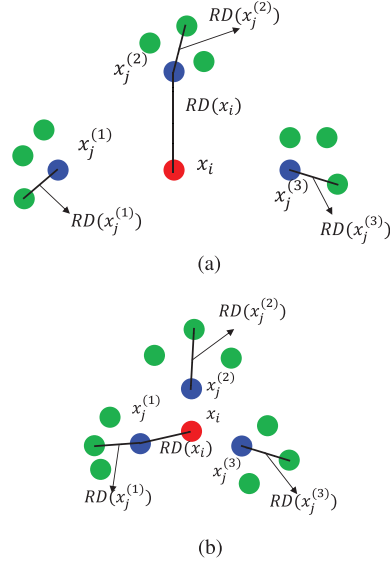


Fig. 2. Visualization of the LOF algorithm with $k = 3$. (a) Example of anomalies ($LOF > 1.5$). (b) Example of normal instance ($LOF \leq 1.5$).

and self-supervised learning-based methods are not effective in handling diverse operating conditions or high-level features. Therefore, a machine learning-based anomaly detection algorithm was adopted. Since healthy data often exhibit multimodal distributions owing to their varying operating conditions, a density-based anomaly detection algorithm is more suitable than other alternatives, such as a one-class support vector machine (OC-SVM), which learns the boundary for healthy data, or iForest, which detects anomalies by partitioning the input data from healthy data. Furthermore, parametric density-based algorithms may also struggle to model such complex data distributions. Based on these considerations, LOF [14], a nonparametric density-based anomaly detection algorithm, was chosen as the anomaly detector.

The LOF algorithm identifies anomalies based on the local density, which represents the density of normal data around an input data sample. Data located in regions with high local density are classified as normal, whereas those with low local density are classified as anomalies. By using the Euclidean distance to measure the distance between two samples, the anomaly detection process for the input data $x \in \mathbb{R}^L$ in a set of normal data samples $X \in \mathbb{R}^{N \times L}$, where the number of samples is N and the data length is L , proceeds as follows.

- 1) Find the k -nearest neighbors of x within X , denoted as $N_k(x; X)$, where k is a user-defined hyperparameter of the LOF algorithm.
- 2) Calculate the k -distance of x as

$$kd(x) = \max_{x_i \in N_k(x; X)} (\|x - x_i\|_2). \quad (8)$$

- 3) Compute the local reachability density of x as

$$LRD(x) = \left(\frac{\sum_{x_i \in N_k(x; X)} RD(x, x_i)}{|N_k(x; X)|} \right)^{-1} \quad (9)$$

where the reachability distance RD between x_i and x_j is defined as

$$RD(x_i, x_j) = \max(kd(x_i), \|x_i - x_j\|_2). \quad (10)$$

- 4) Calculate the LOF of x as

$$LOF(x) = \frac{1}{|N_k(x; X)|} \sum_{x_i \in N_k(x; X)} \frac{LRD(x_i)}{LRD(x)}. \quad (11)$$

- 5) Classify x as an anomaly when $LOF > 1.5$. Otherwise, classify it as normal.

Algorithm 2 Procedure for Anomaly Detection

Require: Input vibration signal $x \in \mathbb{R}^L$

Ensure:

$C \leftarrow$ number of classes, $L \leftarrow$ data length,
 Auxiliary classifier $f(\cdot) : \mathbb{R}^L \rightarrow \mathbb{R}^C$,
 Training dataset $X \in \mathbb{R}^{N_{\text{train}} \times L}$,
 Hybrid feature from training dataset $H \in \mathbb{R}^{N_{\text{train}} \times (C+6)}$,
 Mean and standard deviation of H : $\mu_H, \sigma_H \in \mathbb{R}^{C+6}$,
 Normalized feature $\tilde{H} \in \mathbb{R}^{N_{\text{train}} \times (C+6)} = (H - \mu_H)/\sigma_H$,
 Number of neighbors of LOF algorithm k .

function DETECT(x)

 Compute auxiliary classifier output $z = f(x)$
 Extract statistics $s = [\mu(x), pk(x), rms(x), CF(x), skew(x), kurt(x)]$
 Integrate features $h = [z, s]$
 Normalize feature $\tilde{h} = (h - \mu_H)/\sigma_H$
 $LOF(\tilde{h}) = \frac{1}{|N_k(\tilde{h}; \tilde{H})|} \sum_{h_i \in N_k(\tilde{h}; \tilde{H})} LRD(h_i)/LRD(\tilde{h})$
 if $LOF(\tilde{h}) > 1.5$ **then**
 return *Faulty*
 else
 return *Healthy*
 end if
end function

Fig. 2 illustrates how the LOF value of a given data x_i changes based on the varying positions of its neighbor data when $k = 3$. Algorithm 2 describes the overall anomaly detection process of the proposed method.

IV. PERFORMANCE EVALUATION

This section presents the performance evaluation results of the proposed method. To assess its anomaly detection performance under various machine operating conditions and failure scenarios, the evaluation was conducted on three different datasets. The proposed method was compared with 11 state-of-the-art anomaly detection methods, including machine learning-based, reconstruction, and self-supervised learning-based methods, as listed in Table III. In addition, to demonstrate the practicality of the proposed method in industrial environments, its computational performance on an edge device was evaluated.

A. Data Preparation

To demonstrate the generalization ability and stability of the proposed method, we employed three machine fault datasets:

TABLE III

SUMMARY AND CATEGORIZATION OF ANOMALY DETECTION MODELS FOR PERFORMANCE COMPARISON

Type	Model	Description
Machine Learning	iForest [15]	Conventional machine learning-based anomaly detection methods
	OC-SVM [16]	
	LOF [14]	
Reconstruction Method	DAE [17]	Reconstruction method with encoder-decoder architectures.
	ATF [29]	
	USAD [18]	Reconstruction methods applying adversarial training strategies.
	GANomaly [19]	
Self-Supervised Learning	DSVDD [20]	Methods combining machine learning algorithms with reconstruction methods.
	DASVDD [33]	
	Giri et al. [41]	
Self-Supervised Learning	ICL [39]	Contrastive learning-based anomaly detection method.

USAD: unsupervised anomaly detection on multivariate time-series.

DSVDD: deep SVDD.

the UOS dataset [11], Huazhong University of Science and Technology (HUST) gearbox dataset [43], and Xi'an Jiaotong University and Changxing Sumyoung Technology (XJTU-SY) dataset [44]. The UOS and HUST datasets were used to evaluate the performance of our approach under varying operating conditions. The UOS dataset contains multiple rotating speeds and bearing types, and the HUST dataset includes diverse rotating speeds and motor load conditions. However, these datasets include only the data collected from artificially generated faults. To ensure a comprehensive consideration of real-world industrial scenarios, we additionally used the XJTU-SY dataset, which includes a run-to-failure scenario under accelerated degradation tests.

Vibration signals in the UOS dataset were collected from the testbed operating at six different rotating speeds (600, 800, 1000, 1200, 1400, and 1600 RPMs) and with three bearing types (deep groove ball, cylindrical roller, and tapered roller bearings). For each operating condition, 32 machine condition scenarios are provided, namely, the healthy state, three bearing fault states, seven rotor fault states, and 21 compound fault states, which are combinations of bearing and rotor faults. The dataset also includes data measured with two sampling rates of 8 and 16 kHz. Among them, the 16-kHz data were selected for the performance evaluation.

In the HUST dataset, vibration signals were collected at five rotating speeds (1200, 1500, 1800, 2100, and 2400 RPMs) and five motor loads (0, 0.113, 0.226, 0.339, and 0.452 Nm). The testbed for this dataset consists of a motor, coupler, and gearbox, and the dataset covers two types of gear faults: broken tooth and missing tooth. Vibration signals were sampled at 25.6 kHz, and the data were collected for 10.2 s for each experiment.

Unlike the UOS and HUST datasets, the XJTU-SY dataset contains vibration signals collected until the end of the lifetime of the bearing. Therefore, this dataset facilitates the observation of real-world natural bearing faults. In the dataset, data were recorded at a sampling rate of 25.6 kHz, with a measurement period of one minute and an individual recording duration of 1.28 s. Five run-to-failure experiments were conducted at three rotating speeds (2100, 2250, and 2400 RPMs). Although the type of bearing fault was identified at the end of each experiment, the exact moment when the

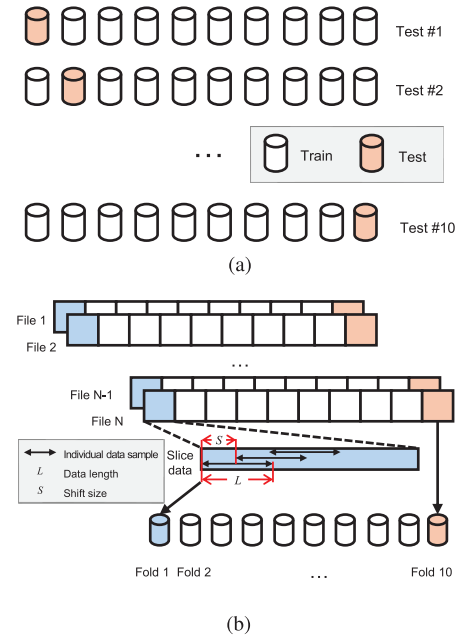


Fig. 3. Data preparation for tenfold cross-validation process. (a) Overview of tenfold cross-validation. (b) Data separation procedure.

initial failure occurred was not specified in the dataset. To estimate this point, the amplitude of the envelope spectrum at each fault frequency was analyzed. The initial fault point was determined as the moment when the amplitude at the fault frequency exceeded the 3σ threshold. Based on this criterion, 30 measurements from the beginning of the data collection were selected as healthy samples, and 30 measurements from the initial fault point were used as faulty samples. Any case containing fewer than 30 measurements of either the healthy or faulty condition was excluded from the experimental analysis.

To ensure the reliability of the experiment, a tenfold cross-validation was conducted. According to the general k -fold cross-validation method, illustrated in Fig. 3(a), the dataset was first divided into tenfold, and then the model was evaluated ten times. In each evaluation, onefold was used for testing, whereas the remaining ninefold was used for training. The three datasets contain data segments corresponding to different operating conditions and fault types. These data segments were first divided into ten subsegments, each assigned to a different fold. The subsegments were further partitioned into smaller data samples with a length L and shift size S . The length L was set to the smallest power of 2 which allowed capturing of at least one full rotation under all operating conditions. The shift size S was initially set equal to the data length L by default. However, in cases where an imbalance occurred between healthy and faulty samples in a dataset, the value of S for a specific health state was reduced to maintain the balance between the number of healthy and faulty samples. The data separation process is illustrated in Fig. 3(b), while Table IV summarizes the values of L , S , and number of samples per fold.

B. Experimental Setup

The model performance was evaluated using accuracy, recall, false positive rate (FPR), precision, $F1$ score, and

TABLE IV

DATA SLICING PARAMETERS AND DISTRIBUTIONS OF THE UOS, HUST, AND XJTU-SY DATASETS

Dataset	Type	Data Length (L)	Shift Size (S)	No. Samples Per Fold
UOS	Healthy	2,048	64	35,442
	Faulty		2,048	34,596
HUST	Healthy	2,048	1,024	600
	Faulty		2,048	600
XJTU-SY	Healthy	1,024	1,024	990
	Faulty		1,024	990

TABLE V

EXPERIMENTAL ENVIRONMENT

Type	Specification
Hardware	CPU Intel Core i9-10900K
	RAM 128GB
	GPU NVIDIA GeForce RTX 2080 SUPER
Software	OS Ubuntu 22.04.4 LTS
	Python 3.12.4
	PyTorch 2.4.0 (with CUDA 11.8)
	scikit-learn 1.5.1

area under the curve (AUC) score. Accuracy represents the proportion of correct predictions among all samples. Although accuracy is one of the most commonly used metrics, the accurate identification of faulty samples is more crucial than the overall accuracy in machine fault diagnosis. Therefore, recall, which indicates the proportion of correctly identified faulty samples among all actual faulty samples, was also evaluated.

However, a strict decision threshold to maximize the recall increases the likelihood of false alarms. Therefore, precision and FPR were reported to evaluate the tradeoff between accurate fault detection and reduction of false alarms. Precision represents the proportion of correctly identified faulty samples among all samples classified as faulty, whereas FPR indicates the proportion of incorrectly identified healthy samples among all actual healthy samples. The $F1$ and AUC scores were also used as evaluation metrics. The $F1$ score, calculated as the harmonic mean of precision and recall, represents the balance between precision and recall. The AUC score is the area under the receiver operating characteristic curve, indicating the threshold-independent model performance.

All metrics range from 0 to 1. A lower FPR value or higher values of the other evaluation metrics indicate improved performance. For the performance evaluation, the average and 95% confidence intervals of these metrics were obtained from tenfold cross-validation.

In the performance evaluation, all methods were trained exclusively on the healthy data from the training folds. For machine learning-based methods, including iForest [15], OC-SVM [16], and LOF [14], the statistical features outlined in Table II were used as the input features, and each method was trained based on its original implementation. In contrast, deep learning-based methods were trained on raw vibration signals for 100 epochs using the Adam optimizer with a

learning rate of 0.001. This setup is based on the benchmark results of Lee and Kim [45], which evaluated optimizers and hyperparameters across various fault diagnosis model architectures.

For the methods using machine learning algorithms, including iForest, OC-SVM, LOF, DSVDD, DASVDD, and the proposed method, the decision thresholds were set to the default values provided by the scikit-learn library. For reconstruction methods, such as DAE, ATF, USAD, and GANomaly, the thresholds were calculated as $Q3 + 1.5IQR$ based on the anomaly score distributions of the training dataset, following the approach outlined in [46]. The details of the experimental environment are summarized in Table V.

C. Performance Comparison Results

The average and 95% confidence intervals of the evaluation metrics for each model are presented in Table VI. In most cases, the proposed approach outperformed all other models used for comparison in this study across all metrics. On the XJTU-SY dataset, OC-SVM achieved a higher recall, and LOF showed better FPR and precision than our method. However, OC-SVM had a much higher FPR, and LOF exhibited a lower recall, indicating that they struggled to balance fault detection accuracy and reduction of false alarms. In contrast, our method achieved the highest accuracy, $F1$ score, and AUC, highlighting its overall effectiveness in anomaly detection.

Among the machine learning-based methods, LOF showed the best anomaly detection performance, supporting the hypothesis that density-based models are well-suited for handling data from diverse operating conditions. Nevertheless, its accuracy and recall remained significantly lower than those of the proposed approach, especially in the case of the XJTU-SY dataset. The reconstruction and self-supervised learning-based methods struggled to maintain consistent performance across various datasets. The performance of the reconstruction methods was significantly lower on the UOS and HUST datasets when compared with that on the XJTU-SY dataset. Conversely, the self-supervised learning-based models exhibited lower performance on the XJTU-SY dataset than on the UOS and HUST datasets.

To further analyze the model performance, the anomaly scores normalized to $[-1, 1]$ in the test dataset and the decision threshold for the healthy and faulty samples are depicted in Fig. 4. Since the tenfold cross-validation results showed minimal performance fluctuations, the results for fold 5 were selected as the representative results. In the figure, the blue vertical line indicates the decision threshold, with samples to the left classified as healthy and those to the right classified as faulty.

The iForest model frequently misclassified faulty samples as healthy. In the case of OC-SVM, the decision threshold was set too low relative to the distribution of healthy data, which resulted in a high number of false positives and, consequently, the lowest precision score among all models. LOF demonstrated a clearer separation between healthy and faulty samples than iForest and OC-SVM. However, it generated more false negatives than our method, which led to a lower recall rate.

TABLE VI
ANOMALY DETECTION PERFORMANCE OF THE PROPOSED METHOD COMPARED WITH STATE-OF-THE-ART METHODS

Dataset	Model	Accuracy	Recall	FPR	Precision	F1	AUC
UOS	iForest	0.837±0.004	0.837±0.002	0.163±0.009	0.834±0.008	0.835±0.004	0.910±0.001
	OC-SVM	0.701±0.008	0.907±0.001	0.501±0.016	0.639±0.007	0.750±0.005	0.898±0.001
	LOF	0.920±0.003	0.856±0.002	0.016±0.006	0.981±0.007	0.914±0.002	0.966±0.002
	DAE	0.799±0.002	0.755±0.003	0.159±0.003	0.823±0.002	0.787±0.002	0.888±0.001
	ATF	0.757±0.002	0.610±0.001	0.100±0.004	0.857±0.005	0.713±0.002	0.831±0.001
	USAD	0.738±0.003	0.560±0.008	0.089±0.007	0.860±0.009	0.678±0.005	0.809±0.003
	GANomaly	0.776±0.007	0.685±0.030	0.135±0.029	0.836±0.026	0.751±0.012	0.820±0.016
	DSVDD	0.758±0.035	0.608±0.074	0.094±0.013	0.861±0.019	0.708±0.058	0.829±0.029
	DASVDD	0.733±0.001	0.566±0.002	0.103±0.003	0.842±0.003	0.677±0.002	0.809±0.001
	Giri et al. [41]	0.822±0.017	0.859±0.010	0.214±0.029	0.797±0.023	0.827±0.015	0.903±0.009
	ICL	0.810±0.004	0.725±0.011	0.106±0.003	0.870±0.003	0.791±0.006	0.851±0.007
	Ours	0.973±0.003	0.956±0.005	0.011±0.005	0.988±0.005	0.972±0.003	0.994±0.001
HUST	iForest	0.723±0.004	0.616±0.006	0.170±0.011	0.784±0.010	0.690±0.004	0.807±0.006
	OC-SVM	0.650±0.016	0.800±0.014	0.500±0.026	0.616±0.014	0.696±0.013	0.780±0.009
	LOF	0.842±0.008	0.709±0.008	0.026±0.011	0.965±0.014	0.817±0.009	0.947±0.006
	DAE	0.772±0.005	0.616±0.006	0.072±0.008	0.896±0.011	0.730±0.005	0.822±0.002
	ATF	0.721±0.002	0.545±0.004	0.104±0.004	0.840±0.005	0.661±0.003	0.790±0.001
	USAD	0.674±0.004	0.378±0.005	0.030±0.006	0.926±0.015	0.537±0.006	0.785±0.001
	GANomaly	0.612±0.052	0.258±0.130	0.034±0.049	0.843±0.215	0.364±0.151	0.704±0.071
	DSVDD	0.699±0.017	0.496±0.037	0.098±0.009	0.834±0.012	0.621±0.031	0.733±0.023
	DASVDD	0.676±0.005	0.381±0.005	0.030±0.007	0.927±0.016	0.540±0.006	0.785±0.001
	Giri et al. [41]	0.876±0.027	0.846±0.032	0.093±0.035	0.902±0.032	0.873±0.026	0.895±0.032
	ICL	0.820±0.006	0.796±0.013	0.156±0.008	0.836±0.007	0.816±0.007	0.885±0.008
	Ours	0.975±0.006	0.967±0.009	0.018±0.010	0.982±0.010	0.974±0.006	0.995±0.002
XJTU-SY	iForest	0.797±0.004	0.777±0.009	0.184±0.004	0.809±0.003	0.793±0.005	0.890±0.002
	OC-SVM	0.712±0.006	0.924±0.005	0.500±0.008	0.649±0.005	0.762±0.005	0.877±0.003
	LOF	0.801±0.003	0.608±0.005	0.005±0.002	0.991±0.003	0.754±0.004	0.912±0.003
	DAE	0.837±0.002	0.781±0.004	0.107±0.004	0.879±0.004	0.827±0.002	0.939±0.001
	ATF	0.827±0.002	0.763±0.003	0.108±0.004	0.876±0.004	0.815±0.002	0.935±0.001
	USAD	0.788±0.002	0.726±0.003	0.149±0.001	0.829±0.002	0.774±0.002	0.918±0.001
	GANomaly	0.806±0.008	0.758±0.033	0.146±0.029	0.842±0.028	0.796±0.012	0.912±0.018
	DSVDD	0.804±0.017	0.721±0.032	0.114±0.003	0.863±0.008	0.785±0.022	0.842±0.021
	DASVDD	0.785±0.001	0.720±0.003	0.150±0.001	0.827±0.001	0.770±0.002	0.919±0.001
	Giri et al. [41]	0.756±0.024	0.689±0.044	0.178±0.007	0.794±0.014	0.737±0.031	0.834±0.032
	ICL	0.674±0.017	0.456±0.032	0.109±0.011	0.807±0.020	0.582±0.029	0.657±0.025
	Ours	0.918±0.006	0.844±0.013	0.008±0.002	0.990±0.002	0.911±0.008	0.977±0.002

TABLE VII
INFERENCE TIME MEASUREMENT RESULTS ON RASPBERRY PI 4B

Procedure	Computation Time (ms)			
	avg.	min.	max.	std.
Statistical Feature Extraction	0.48	0.47	0.57	0.02
Auxiliary Classifier Inference	2.58	2.48	3.51	0.13
Anomaly Detection (LOF)	2.80	2.68	3.50	0.13
Total	5.88	5.74	6.77	0.17

In the UOS and HUST datasets, the anomaly scores of healthy samples derived from most reconstruction methods, except for DSVDD, exhibited a multimodal distribution with multiple peaks. Furthermore, some distributions extended beyond the decision threshold, which led to performance degradation. This suggests that the low performance of reconstruction methods is primarily due to the complexity of the healthy data distribution. Since DSVDD uses the features from a neural network trained to project normal data into a hyperspherical region, the anomaly scores of healthy samples in this method formed an unimodal distribution. However, many faulty samples were assigned anomaly scores below the decision threshold, indicating that these models struggled to effectively distinguish between healthy and faulty samples.

Unlike the reconstruction methods, the anomaly scores of healthy samples generated by self-supervised learning-based methods formed an unimodal distribution across all datasets. They outperformed the reconstruction methods on the UOS and HUST datasets. However, in the XJTU-SY dataset, the anomaly scores of many faulty samples were below the decision threshold, leading to a significantly lower recall rate and degradation of the overall anomaly detection performance. This result underscores their limitations in accurately classifying the natural machine fault in the initial stages. In contrast, our method consistently achieved superior anomaly detection performance across all datasets.

D. Real-Time Performance Evaluation

In a real-world industrial setup, sensor data are acquired from edge devices with limited hardware resources. Moreover, fault detection plays a crucial role in ensuring the safe operation of mechanical systems. Therefore, achieving real-time responsiveness on resource-constrained edge devices is essential for the practical deployment of an anomaly detection method. To assess the real-time performance of our method, we measured the inference time of the model on a Raspberry Pi 4B.

Table VII presents the average, minimum, maximum, and standard deviation of the inference time measured over 100

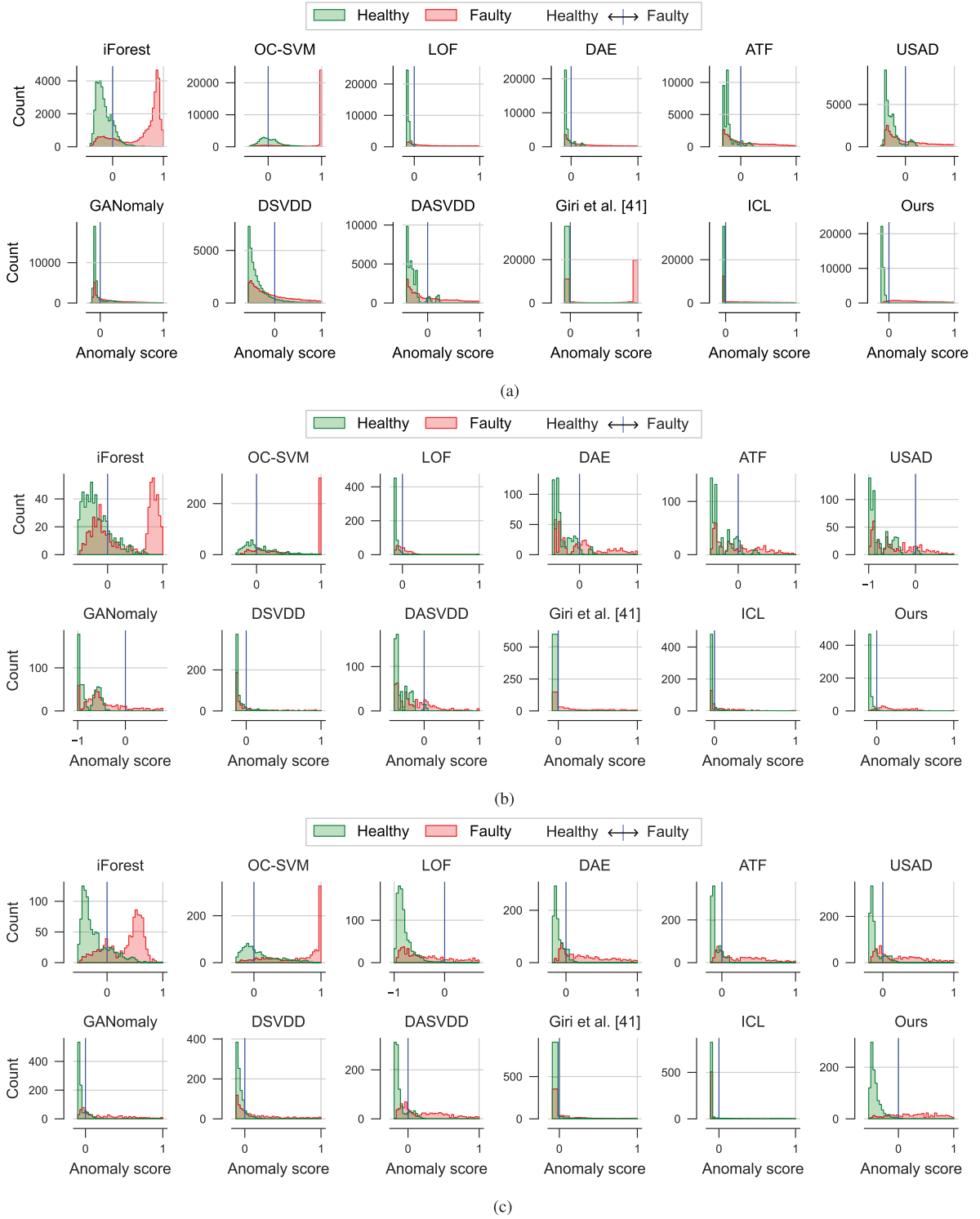


Fig. 4. Visualization of decision threshold and anomaly score distribution of healthy and faulty samples for comparison models. (a) UOS dataset. (b) HUST dataset. (c) XJTU-SY dataset.

computations after 1000 warm-up runs. The results showed that our method completed the anomaly detection process in a maximum of 6.77 ms, even without additional model optimizations. This is a sufficiently short time when compared with the data measurement durations of the datasets: 128, 80, and 40 ms for the UOS, HUST, and XJTU-SY datasets, respectively. Consequently, our method can perform real-time

anomaly detection on edge devices before the next data sample is collected.

V. ANALYSIS OF THE PROPOSED METHOD

This section presents two analyses to demonstrate the advantages of the proposed hybrid feature. First, the performance of the hybrid feature was compared with that of alternative

TABLE VIII
RESULTS OF ABLATION STUDIES ON INPUT FEATURES OF THE LOF ALGORITHM

Dataset	Features	Accuracy	Recall	FPR	Precision	F1	AUC
UOS	Stats.	0.920±0.003	0.856±0.002	0.016±0.006	0.981±0.007	0.914±0.002	0.966±0.002
	Aux.	0.910±0.014	0.832±0.030	0.014±0.003	0.983±0.004	0.901±0.017	0.971±0.005
	Stats. + Rot.	0.946±0.003	0.909±0.001	0.017±0.006	0.981±0.006	0.944±0.003	0.979±0.001
	Stats. + Aux. (Ours)	0.973±0.003	0.956±0.005	0.011±0.005	0.988±0.005	0.972±0.003	0.994±0.001
HUST	Stats.	0.842±0.008	0.709±0.008	0.026±0.011	0.965±0.014	0.817±0.009	0.947±0.006
	Aux.	0.968±0.013	0.963±0.022	0.026±0.013	0.974±0.012	0.968±0.014	0.989±0.007
	Stats. + Rot.	0.849±0.004	0.713±0.006	0.016±0.004	0.979±0.006	0.825±0.005	0.953±0.004
	Stats. + Aux. (Ours)	0.975±0.006	0.967±0.009	0.018±0.010	0.982±0.010	0.974±0.006	0.995±0.002
XJTU-SY	Stats.	0.801±0.003	0.608±0.005	0.005±0.002	0.991±0.003	0.754±0.004	0.912±0.003
	Aux.	0.678±0.023	0.375±0.046	0.018±0.003	0.951±0.010	0.535±0.051	0.771±0.034
	Stats. + Rot.	0.883±0.004	0.776±0.007	0.010±0.003	0.987±0.003	0.869±0.005	0.959±0.003
	Stats. + Aux. (Ours)	0.918±0.006	0.844±0.013	0.008±0.002	0.990±0.002	0.911±0.008	0.977±0.002

Stats.: statistical features, Aux.: auxiliary classifier outputs, Rot.: rotating speed.

feature sets through ablation studies. Next, its effectiveness was evaluated by visualizing the statistical features, auxiliary classifier outputs, and the hybrid feature. All experiments in this section were conducted using the same setup described in Section IV.

A. Ablation Studies

The performance of the hybrid feature was compared with those of the following three alternative feature sets: statistical features (Stats.), auxiliary classifier outputs (Aux.), and the combination of the Stats. and rotating speed (Stats. + Rot.). These alternative feature sets were derived by either omitting or substituting specific components of the hybrid feature. To ensure a fair comparison, the LOF algorithm was used as the anomaly detector across all feature sets.

Table VIII presents the comparison results between the feature sets. Our proposed hybrid feature consistently outperformed all alternative feature sets in most cases. In contrast, the alternative feature sets exhibited inferior performance, particularly in terms of recall, which directly affected the overall anomaly detection accuracy. Although the Stats. + Rot. feature showed a slightly higher FPR on the HUST dataset and the Stats. feature achieved better FPR and precision on the XJTU-SY dataset, these performance gains were negligible compared to the substantial improvements provided by the hybrid feature.

Moreover, the hybrid feature led to significantly higher accuracy, $F1$ score, and AUC, confirming its superiority in overall performance. The ablation study results highlight that the hybrid feature is the most effective combination, significantly enhancing the anomaly detection performance when compared with the other feature sets.

B. Feature Visualization Analysis

To further understand the performance enhancement attributable to the hybrid feature in the proposed method, this section analyzes the fault-discriminative ability of the individual and hybrid features through visualization techniques. First, the statistical features and auxiliary classifier outputs were analyzed using histograms to understand their

individual contributions to the fault discrimination. Second, to demonstrate the effectiveness of the hybrid feature for fault discrimination, we visualized and compared the statistical features, auxiliary classifier outputs, and the proposed hybrid feature in the latent feature space.

To demonstrate the fault-discriminative ability of the individual features, histograms of the features derived from the statistical parameters and auxiliary classifier are illustrated in Figs. 5 and 6, respectively. Fig. 5 shows the histograms of six statistical features for healthy and faulty data. The distributions of μ and skew showed minimal differences between faulty and healthy samples across all datasets. In contrast, the values of pk and rms were consistently higher in faulty data across all datasets, and CF and kurt served as fault-discriminative indicators in the UOS and HUST datasets. This indicates that machine faults in the datasets typically amplify the magnitude and frequency of extreme values in the vibration signals, enabling the statistical features to differentiate faulty data from healthy data. However, because the signal shapes are affected not only by machine faults but also by the operating conditions, overlaps between the features of healthy and faulty data were observed. Therefore, relying solely on statistical features is insufficient for distinguishing between healthy and faulty data.

The outputs of the auxiliary classifier, which was trained to classify rotating speeds, exhibited distinct distributions for different rotating speeds. Fig. 6(a)–(c) presents the histograms of the auxiliary classifier outputs for the UOS, HUST, and XJTU-SY datasets collected at 1600, 1200, and 2250 RPMs, respectively. As described in Section III-A, each z_i value in the figures represents the logit for the i th rotating speed class. For the healthy data, the outputs corresponding to the actual rotating speed class were the highest among all class outputs, typically greater than 0, whereas most outputs for the other classes were below 0. Conversely, because faulty data were not included in the training dataset, their outputs were noticeably different from those of the healthy samples.

These results reveal that the classifier outputs derived from faulty data exhibit patterns distinct from those of outputs derived from healthy data. This pattern remains consistent

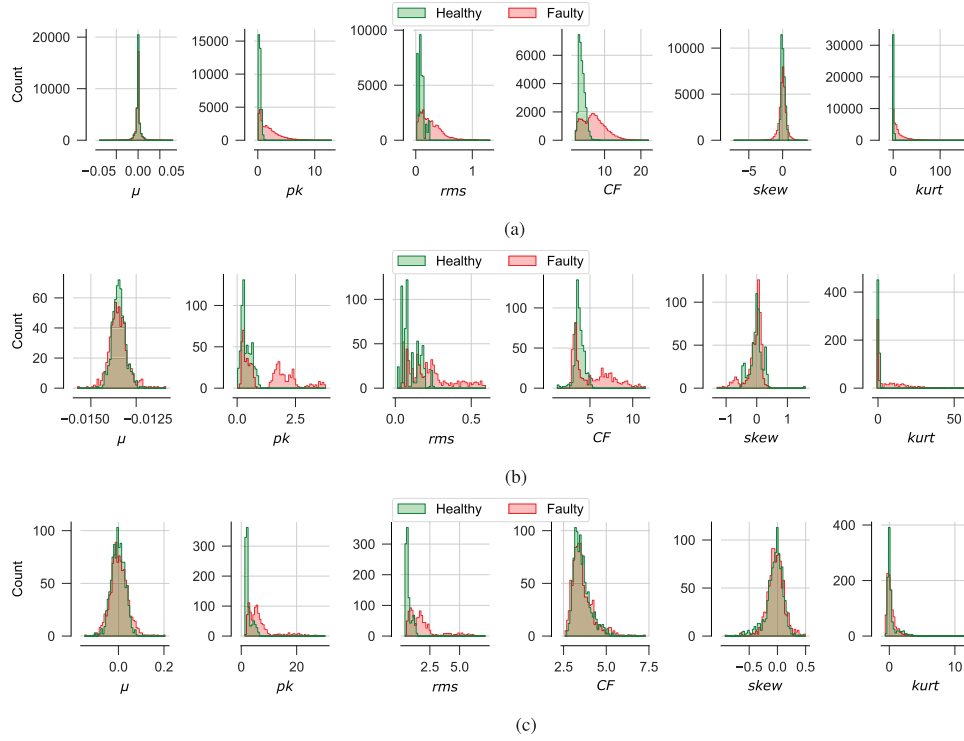


Fig. 5. Comparison of statistical feature distributions between healthy and faulty samples. (a) UOS dataset. (b) HUST dataset. (c) XJTU-SY dataset.

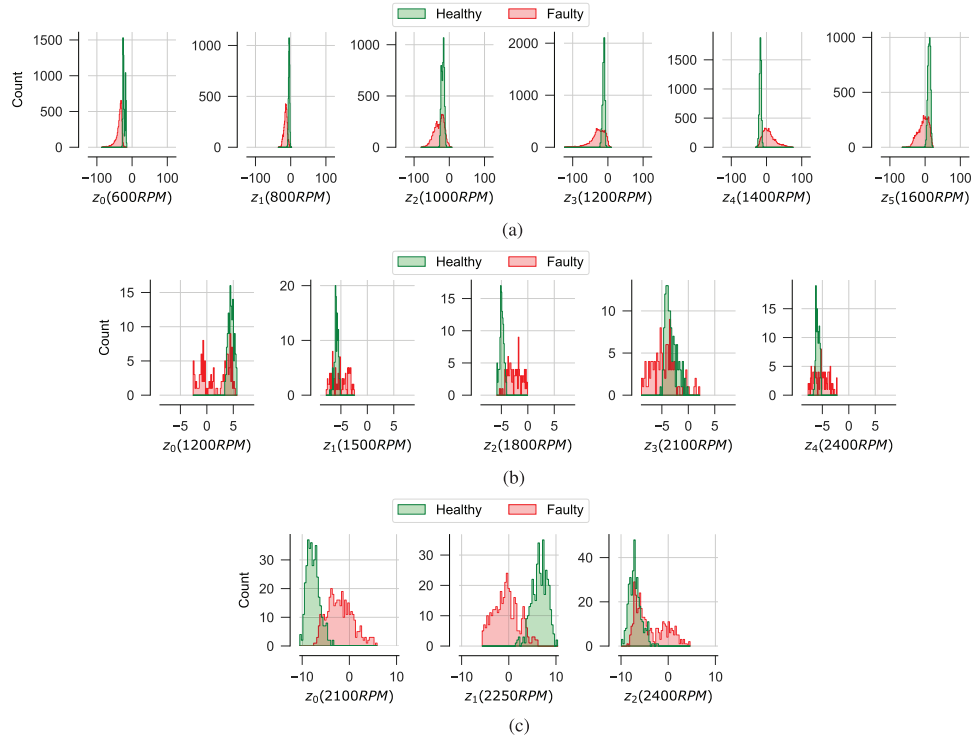


Fig. 6. Comparison of auxiliary classifier output distributions between healthy and faulty samples. (a) UOS dataset (1600 RPM data). (b) HUST dataset (1200 RPM data). (c) XJTU-SY dataset (2250 RPM data).

across all rotating speeds and datasets, indicating that the auxiliary classifier outputs can effectively separate faulty data from healthy data. However, some faulty data samples exhibited feature patterns similar to those of the healthy data samples, causing partial overlaps in the histograms shown in Fig. 6. As a result, although both statistical features and

auxiliary classifier outputs are useful for machine anomaly detection, neither feature by itself is sufficient to clearly distinguish faulty samples from healthy ones under diverse operating conditions.

To demonstrate the effectiveness of the hybrid feature for fault discrimination, we compared the statistical features,

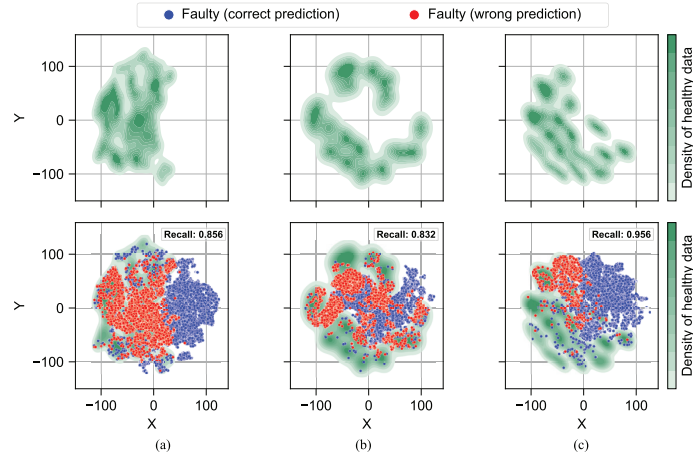


Fig. 7. Comparison of features for fault discrimination capabilities using t-SNE visualization (UOS dataset). (a) Statistical features. (b) Auxiliary classifier outputs. (c) Hybrid feature (proposed).

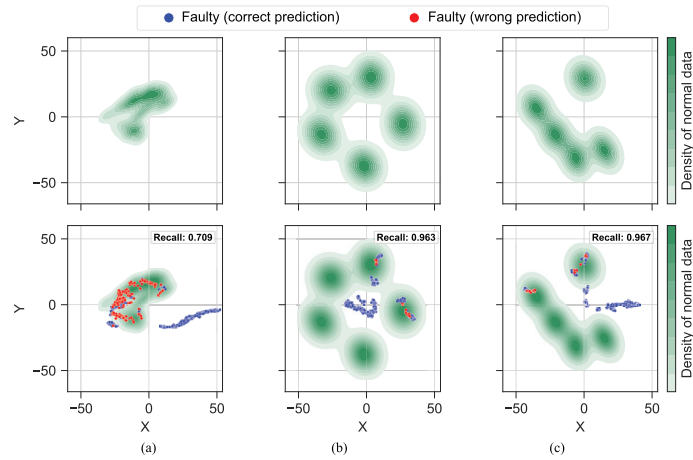


Fig. 8. Comparison of features for fault discrimination capabilities using t-SNE visualization (HUST dataset). (a) Statistical features. (b) Auxiliary classifier outputs. (c) Hybrid feature (proposed).

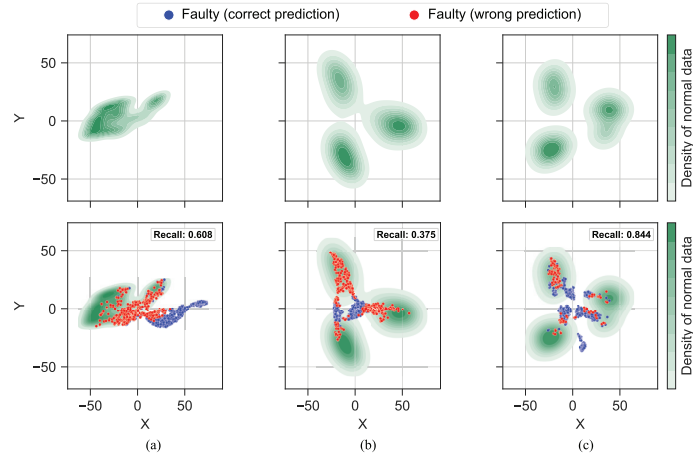


Fig. 9. Comparison of features for fault discrimination capabilities using t-SNE visualization (XJTU-SY dataset). (a) Statistical features. (b) Auxiliary classifier outputs. (c) Hybrid feature (proposed).

auxiliary classifier outputs, and the proposed hybrid feature in the latent feature space. Figs. 7–9 illustrate the 2-D projections of these three features using t-distributed stochastic neighbor embedding (t-SNE) on the UOS, HUST, and XJTU-SY datasets, respectively. The t-SNE algorithm retains the

local structure of the data while reducing their dimensionality. Therefore, high-dimensional data characteristics can be effectively visualized in a lower-dimensional space. In the figures, the first row presents the density maps of each feature derived from the training dataset. In the second row, the detection

results of faulty samples in the test dataset generated by the LOF algorithm are overlaid on the corresponding density maps. Because the LOF algorithm relies on the local density of the training data to detect anomalies, samples located in sparser regions are more likely to be identified as faulty. Blue dots indicate the correctly predicted faulty samples, whereas red dots represent the misclassified faulty samples.

The t-SNE visualization results showed that when individual features were used, many faulty samples were mapped into the high-density regions of the healthy data, leading to a low recall rate. For example, as shown in Figs. 8(a) and 9(b), using the statistical features in the HUST dataset and the auxiliary classifier outputs in the XJTU-SY dataset resulted in significantly lower recall rates than that observed when using the hybrid feature. By contrast, as illustrated in Figs. 7(c), 8(c), and 9(c), the hybrid feature provides much clearer separation between healthy and faulty samples across all datasets, leading to the highest recall rate. These results suggest that the magnitude and shape information from the statistical features and the operating condition-related fault characteristics captured by the auxiliary classifier outputs complement each other in effectively representing the characteristics of machine faults.

VI. CONCLUSION AND FUTURE WORK

This study proposes a novel machine anomaly detection method to overcome two key challenges in deploying deep learning-based fault diagnosis for rotating machinery in real-world industrial settings: the scarcity of labeled faulty data and the variability of machine operating conditions. To overcome these challenges, we first designed a pretext task to classify rotating speeds measured by Hall sensors and trained the auxiliary classifier through this task. Then, based on the auxiliary classifier outputs, we proposed a hybrid feature integration approach that combines them with vibration statistics.

To evaluate our method under diverse machine operating conditions and fault types, experiments were conducted on the following three datasets: UOS, HUST, and XJTU-SY. The proposed method consistently outperformed 11 comparison models across all datasets and enabled real-time anomaly detection on a Raspberry Pi 4B. Moreover, extensive ablation studies and feature visualizations confirmed the effectiveness of the hybrid feature. The analysis revealed that the hybrid feature effectively compensates for the limitations of individual statistical features or auxiliary classifier outputs, enabling a clear separation between healthy and faulty data.

In future work, we aim to develop a more robust anomaly detection algorithm capable of handling extended machine operating conditions, limited data availability, and constrained hardware resources for efficient unsupervised on-device fault diagnosis.

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